



Team Design Critique Presentation

Team 2 – *Mathified*

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What is Mathified?

Mathified is a learning platform designed to help students from Grade 6 to College.

It helps learners practice math through games that they can interact with to make the learning experience interactive and fun.

Mathified provides a different medium to reward users for solving problems and offers help to those who seek it.



Overview of Usability

1. Visibility of System Status

-  Real-time progress updates
-  No feedback after clicking "Game mode"
-  **Fix:** Added more difficulty

2. Match Between System and Real World

-  Familiar icons (e.g., calculator)
-  Confusing terminology
-  **Fix:** Simplified wording

3. User Control and Freedom

-  Navigation was restricted
-  **Fix:** Maybe more bottom to jump back to main page



Overview of Usability

4. Consistency and Standards

- ❌ Icons like the lightbulb had multiple meanings
- 🔧 **Fix:** Reused icons consistently (lightbulb only = hint)

5. Error Prevention

- ❌ Lightbulb button bot function
- 🔧 **Fix:** Added validation and disabled premature submission

6. Recognition Rather Than Recall

- ❌ Lightbulb Buttons were scattered
- 🔧 **Fix:** Standardized layout and flow



Overview of Usability

7. Flexibility and Efficiency

-  Users had to click through every step
-  **Fix:** Jump navigation introduced

8. Aesthetic and Minimalist Design

-  No games picture
-  **Fix:** Add some more pictures or icons to make it more colorful

9. Help Users Recognize, Diagnose, and Recover from Errors

-  Errors weren't visible
-  **Fix:** Clear error messages and visual cues

10. Help and Documentation

-  Comprehensive in-app help and README
- No action needed



Design Improvement Recommendations

- **Accessibility Enhancements**
 - High-contrast theme option
 - Larger fonts, keyboard-friendly navigation
 - Improves usability for all students
- **Expand Math Content**
 - Add more branches: Pre-Algebra, Algebra 2, Trigonometry
 - Complete detailed content for each branch
 - Supports a wider learning range
- **Deeper Gamification**
 - Add XP system, badges, achievements
 - Reward users for milestones
 - Encourages repeat engagement



Design Improvement Recommendations

- **User Profile & Progress Tracking**
 - Dashboard to view completed topics, scores, and history
 - Personalized learning progress overview
 - Motivates long-term use
- **Mobile Responsiveness**
 - Optimize layout for phones and tablets
 - Improve touch controls and responsiveness



Wrap Up

- **Strong Foundation:**
 - Mathified makes math learning more engaging through games.
- **Key Strengths:**
 - Clear navigation, intuitive feedback, and user-friendly design.
- **Opportunities for Improvement:**
 - Add accessibility options
 - Expand math content and game variety
 - Enhance game flow and user personalization
- **Overall:**
 - A solid platform with great potential to grow into a powerful, inclusive learning tool.